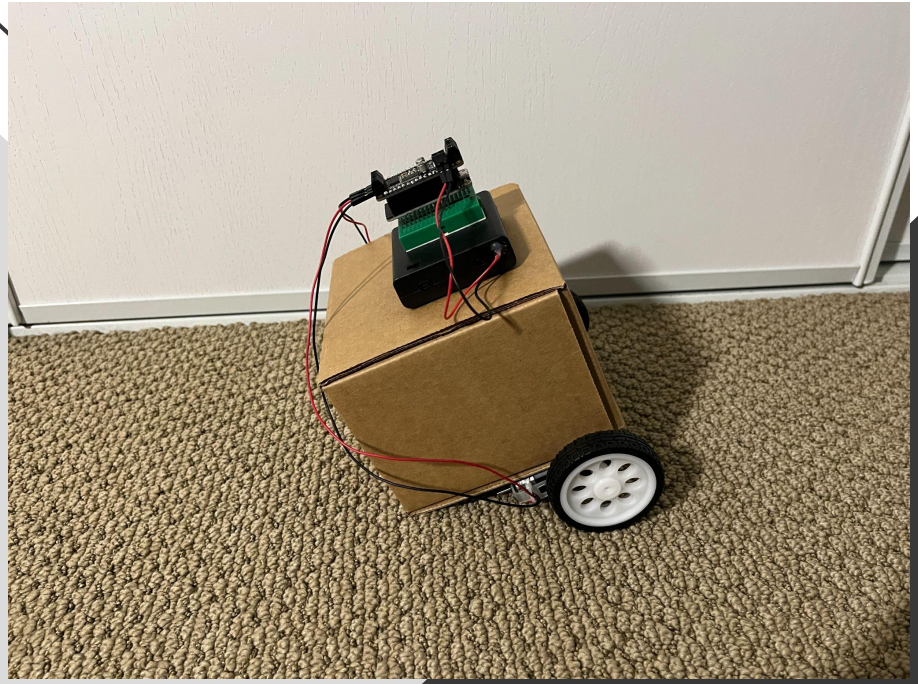


Final Project: Bluetooth Controlled Vehicle

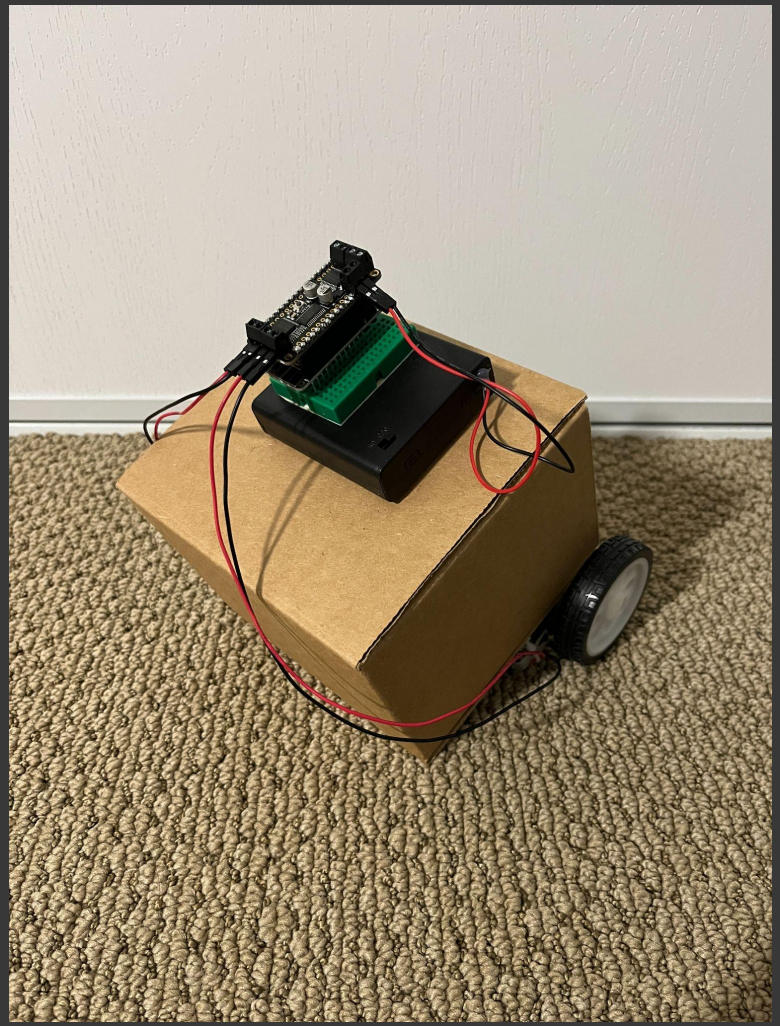
DES INV 22: FALL 2022

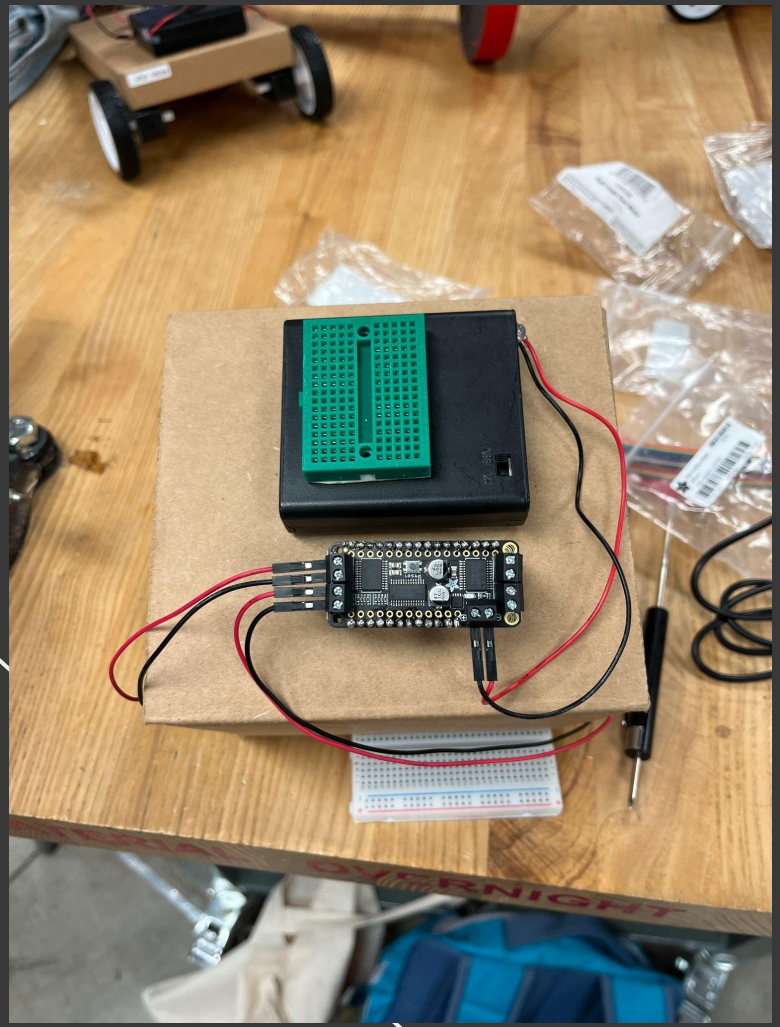
Christy Quang

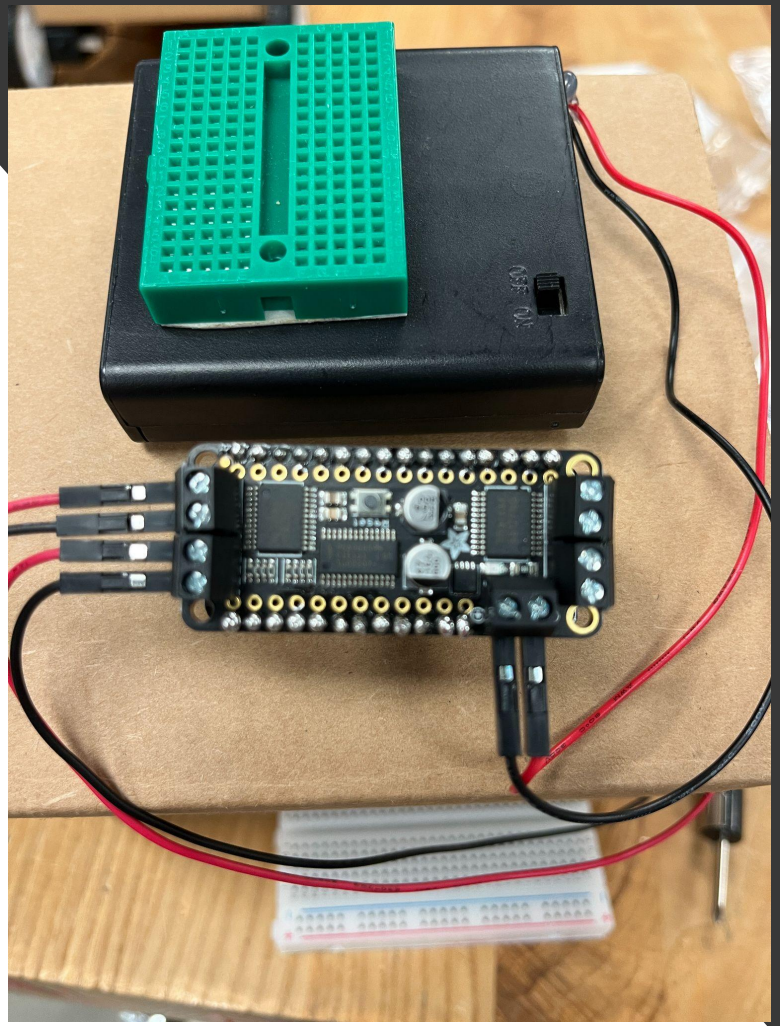
PRODUCT DESCRIPTION

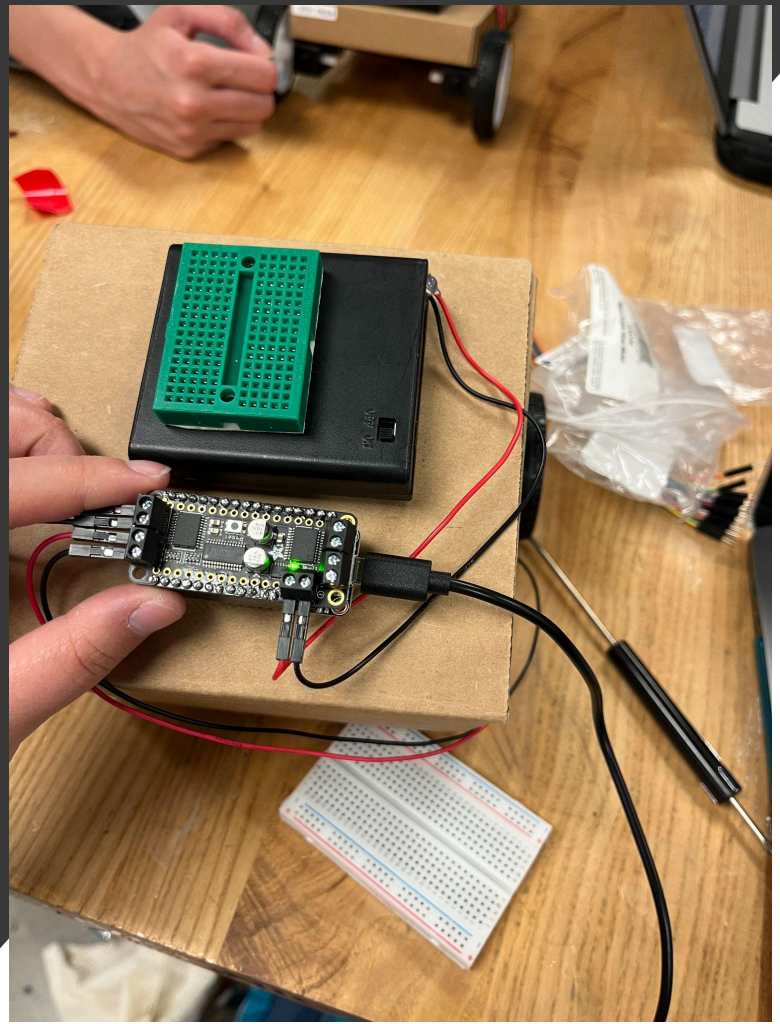
Simple, bluetooth controlled two-wheeled car.

Video Link: https://youtu.be/2rm_lgl--d0






OFF ON



No changes to the code

Ada_BLE_RC | Arduino 1.8.19

Ada_BLE_RC | BluefruitConfig.h | packetParser.cpp

```
// Check for originalMin > originalMax  - the math for all other cases i.e. negative numbers seems to work out fine
if (originalMin > originalMax ) {
  return 0;
}

if (invFlag == 0){
```

Error compiling for board Adafruit ESP32 Feather.

Copy error messages

```
sketch/packetParser 3.cpp.o: In function `parsefloat(unsigned char*)':
/Users/christyquang/Ada_BLE_RC/packetParser 3.cpp:34: multiple definition of `parsefloat(unsigned char*)'
sketch/packetParser 2.cpp.o:/Users/christyquang/Ada_BLE_RC/packetParser 2.cpp:34: first defined here
sketch/packetParser 3.cpp.o: In function `printHex(unsigned char const*, unsigned int)':
packetParser 3.cpp:(.text._Z8printHexPKhj+0x0): multiple definition of `printHex(unsigned char const*, unsigned int)'
sketch/packetParser 2.cpp.o:packetParser 2.cpp:(.text._Z8printHexPKhj+0x0): first defined here
sketch/packetParser 3.cpp.o:(.bss.packetbuffer+0x0): multiple definition of `packetbuffer'
sketch/packetParser 2.cpp.o:(.bss.packetbuffer+0x0): first defined here
sketch/packetParser 3.cpp.o: In function `readPacket(Adafruit_BLE*, unsigned short)':
packetParser 3.cpp:(.text._Z10readPacketP12Adafruit_BLEt+0x0): multiple definition of `readPacket(Adafruit_BLE*, unsigned short)'
sketch/packetParser 2.cpp.o:packetParser 2.cpp:(.text._Z10readPacketP12Adafruit_BLEt+0x0): first defined here
sketch/packetParser.cpp.o: In function `parsefloat(unsigned char*)':
/Users/christyquang/Ada_BLE_RC/packetParser.cpp:34: multiple definition of `parsefloat(unsigned char*)'
sketch/packetParser 2.cpp.o:packetParser 2.cpp:(.text._Z10parsefloatPh+0x0): first defined here
sketch/packetParser.cpp.o: In function `printHex(unsigned char const*, unsigned int)':
packetParser.cpp:(.text._Z8printHexPKhj+0x0): multiple definition of `printHex(unsigned char const*, unsigned int)'
sketch/packetParser 2.cpp.o:packetParser 2.cpp:(.text._Z8printHexPKhj+0x0): first defined here
sketch/packetParser.cpp.o:(.bss.packetbuffer+0x0): multiple definition of `packetbuffer'
sketch/packetParser 2.cpp.o:(.bss.packetbuffer+0x0): first defined here
sketch/packetParser.cpp.o: In function `readPacket(Adafruit_BLE*, unsigned short)':
packetParser.cpp:(.text._Z10readPacketP12Adafruit_BLEt+0x0): multiple definition of `readPacket(Adafruit_BLE*, unsigned short)'
sketch/packetParser 2.cpp.o:packetParser 2.cpp:(.text._Z10readPacketP12Adafruit_BLEt+0x0): first defined here
collect2: error: ld returned 1 exit status
exit status 1
Error compiling for board Adafruit ESP32 Feather.
```

1

Adafruit ESP32 Feather, 80MHz, 230400, None, Default on /dev/cu.usbserial-02650266

PROCESS

The first step is finding a small cardboard box that would serve as the bluetooth controlled vehicle. After soldering the Arduino and Adafruit bluetooth boards together, the slideshow presentation provided made setting up the car fairly easy.

However, setting up the code was very difficult since there were instructions that weren't explicitly mentioned on the Adafruit website. For instance, all of the files needed to be put in the same project directory in addition to supplementary libraries.

While the code compiled and was sent to my board, the vehicle was unable to start and actually move, and the class wasn't able to get help since Professor Myers is sick.

Bill of Materials

- 2 - Wheels
- 2 - Right angle motors
- Double sided tape
- Small cardboard box
- Battery box
- 4 AA batteries
- Arduino
- Adafruit BLE
- Micro-usb to USB-C converter

REFLECTION

SURPRISES ENCOUNTERED

The code was hard to troubleshoot – the setup was unintuitive and there were a lot of debugging issues that were annoying to deal with.

FRUSTRATING PARTS

The Arduino bluetooth code was the most frustrating part because I followed the instructions but there were hidden ones that made the entire set up difficult. There was also no one to help during class so all of the students were all struggling together.

DIFFERENT APPROACHES

I would have chosen a smaller box so the wires can be kept more compact. I would have also started this project earlier (last week) so I could get all of the necessary supplies and help.

THANK YOU!